

ICC-Tutorial: Worksheet for an MILP encoding of Neural Networks

Lipsy Gupta¹, Ratan Lal², Pavithra Prabhakar³

¹Kansas State University

²Northwest Missouri State University

³University of New Mexico

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1 Worked Example: MILP Encoding of a Two-Layer NN

Goal

- Encode a closed-loop system $S = (f, \mathcal{N})$ into logical (MILP) constraints.
- Separate encoding for:
 - Neural network controller $\mathcal{N}$,
 - Plant dynamics f ,
 - Executions (trajectories) of the closed-loop system.
- Use Big- M constraints to encode ReLU activations.

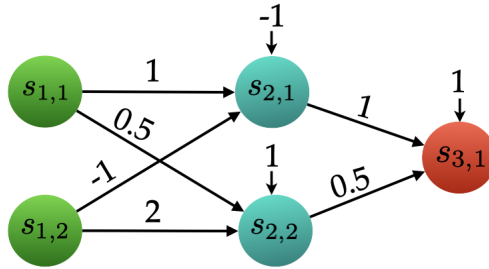


Figure 1: Neural Network $\mathcal{N}_1$

We consider the concrete two-layer network $\mathcal{N}_1 = \{(W_1, b_1), (W_2, b_2)\}$ shown in Fig. 1 and illustrate how its behavior is encoded using linear constraints.

Variables For each node s_{ij} , we denote the value at the node by the variable x_{ij} .

- Input layer: x_{11}, x_{12}
- Hidden layer: x_{21}, x_{22}
- Output layer: x_{31}

The variable names associated with each neuron are shown in Fig. 2.

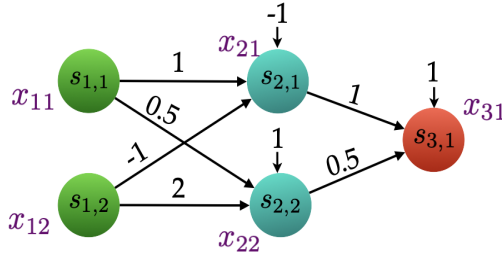


Figure 2: Neural Network $\mathcal{N}_1$

Encoding for the neuron $s_{1,1}$: We write the constraints for the value $x_{2,1}$. Note that

$$x_{21} = \text{ReLU}(1 \cdot x_{11} + (-1) \cdot x_{12} + (-1)),$$

Let

$$y_{21} = 1 \cdot x_{11} + (-1) \cdot x_{12} + (-1)$$

Now, the value of x_{21} is given by

$$x_{21} = \text{ReLU}(y_{21}),$$

i.e.,

if $y_{21} \geq 0$, then $x_{21} = y_{21}$; else $x_{21} = 0$.
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The above definition of x_{21} uses a conditional statement (“if-then-else”), which is not allowed in a linear program. To encode this behavior using linear constraints, we introduce a binary variable that indicates whether the condition $y_{21} \geq 0$ is active.

Let q_{21} be a binary variable, i.e., it takes either the value 0 or 1. The variable q_{21} is intended to be 1 when $y_{21} \geq 0$, and 0 otherwise. Using a sufficiently large constant $M > 0$, the conditional definition of x_{21} can be enforced using the following linear constraints.

Constraints:

$$y_{21} \leq x_{21}, \quad (1)$$

$$y_{21} + Mq_{21} \geq x_{21}, \quad (2)$$

$$x_{21} \geq 0, \quad (3)$$

$$x_{21} \leq M(1 - q_{21}). \quad (4)$$

How the Big- M encoding enforces the definition of x_{21} :

- **Case $y_{21} > 0$:**
 - From constraint (1), we have $x_{21} \geq y_{21} > 0$, i.e., $x_{21} > 0$.
 - If $q_{21} = 1$, then constraint (4) gives $x_{21} \leq 0$, which is contradiction to $x_{21} > 0$, hence $q_{21} = 0$. This makes $x_{21} \leq M$, which is true.
 - With $q_{21} = 0$, constraint (2) becomes $x_{21} \leq y_{21}$.
 - Combined with constraint (1), we obtain $x_{21} = y_{21}$.
 - Note that the only remaining constraint (3) given by $x_{21} \geq 0$, is also satisfied
 - We get $x_{21} \leq y_{21}$ as desired.
- **Case $y_{21} < 0$:**
 - Constraint (3) enforces $x_{21} \geq 0$.
 - Constraint (1) gives $x_{21} \geq y_{21}$, which is automatically satisfied since $y_{21} < 0$.
 - If $q_{21} = 0$, constraint (2) requires $x_{21} \leq y_{21} < 0$, which contradicts $x_{21} \geq 0$.
 - Hence, $q_{21} = 1$.
 - With $q_{21} = 1$, constraint (4) gives $x_{21} \leq 0$.
 - Together with $x_{21} \geq 0$, this yields $x_{21} = 0$.
- **Case $y_{21} = 0$:**
 - Similar proof, check yourself

Encoding all nodes. We repeat the same process for every neuron. Concretely, for each value x_{ij} of such neuron, we:

- have affine transformation variable y_{ij} ,
- have a binary variable q_{ij} ,
- use a sufficiently large constant $M > 0$.

Encoding for the neuron $s_{1,2}$: We write the constraints for the hidden neuron $s_{1,2}$ - value denoted by x_{22} . Let

$$y_{22} = 0.5 \cdot x_{11} + 2 \cdot x_{12} + 1$$

Now, the value of x_{22} is given by

$$x_{22} = \text{ReLU}(y_{22}).$$

Using a binary variable q_{22} and the constant $M > 0$, we add:

$$y_{22} \leq x_{22}, \tag{5}$$

$$y_{22} + Mq_{22} \geq x_{22}, \tag{6}$$

$$x_{22} \geq 0, \tag{7}$$

$$x_{22} \leq M(1 - q_{22}). \tag{8}$$

Note on the choice of M . The constant M can be chosen the same for all neurons, but for more efficient computation it is common to use a different (tighter) value of M for each neuron. In this worked example, we use the following Big- M bounds:

$$M_{21} = 5.0, \quad M_{22} = 6.0, \quad M_{31} = 9.0.$$

2 Exercise

Exercise 1 (output neuron $s_{3,1}$). Write the encoding for the value x_{31} at this output neuron :

- First define its affine transformation y_{31} in terms of x_{21}, x_{22} using the weights and bias from the network in Fig. 1.
- Then write the Big- M constraints for $x_{31} = \text{ReLU}(y_{31})$ by introducing a binary variable q_{31} .
- Use $M_{31} = 9.0$.

MILP constraints for $\mathcal{N}_1$

Step 2: Encoding the Full Neural Network Controller

$$\left\{ \begin{array}{l} y_{21} = 1x_{11} - 1x_{12} - 1 \\ y_{21} \leq x_{21} \\ y_{21} + 5.0 q_{21} \geq x_{21} \\ x_{21} \geq 0 \\ x_{21} \leq 5.0(1 - q_{21}) \end{array} \right. \quad (s_{2,1})$$

$$\left\{ \begin{array}{l} y_{22} = 0.5x_{11} + 2x_{12} + 1 \\ y_{22} \leq x_{22} \\ y_{22} + 6.0 q_{22} \geq x_{22} \\ x_{22} \geq 0 \\ x_{22} \leq 6.0(1 - q_{22}) \end{array} \right. \quad (s_{2,2})$$

$$\left\{ \begin{array}{l} y_{31} = 1x_{21} + 0.5x_{22} + 1 \\ y_{31} \leq x_{31} \\ y_{31} + 9.0 q_{31} \geq x_{31} \\ x_{31} \geq 0 \\ x_{31} \leq 9.0(1 - q_{31}) \end{array} \right. \quad (s_{3,1})$$

$$q_{21}, q_{22}, q_{31} \in \{0, 1\}.$$

Exercise 2 (updated network). Consider the updated network shown in Fig. 3.

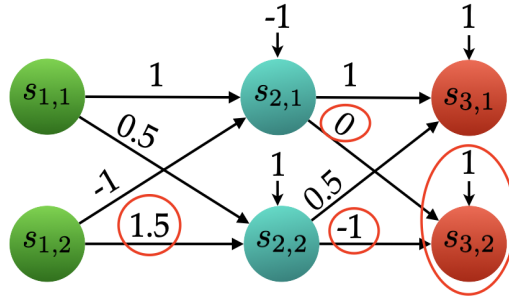


Figure 3: Updated neural network

Note. In the updated network: (i) the weight from $s_{1,1}$ to $s_{2,2}$ is changed to 1.5, and (ii) a second output neuron $s_{3,2}$ is introduced.

For this network, use the following Big- M bounds:

$$M_{21} = 5.0, \quad M_{22} = 5.0, \quad M_{31} = 8.5, \quad M_{32} = 6.0.$$

- Write encoding for value at $s_{2,2}$.
- Write encoding for value at $s_{3,2}$.

Step 3: Encoding Plant Dynamics

Variables:

- $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$: current state.
- u : control input.
- $x' = \begin{bmatrix} x'_1 \\ x'_2 \end{bmatrix}$: next state.

Linear dynamics:

$$x(t+1) = Ax(t) + Bu(t),$$

Using

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

we obtain the constraints:

$$\begin{cases} x'_1 = x_1 + x_2 \\ x'_2 = -x_1 + x_2 + u \end{cases}$$

Note that u is nothing but NN output (x_{31} from previous encoding). The state x' becomes the input to the NN at the next step. Hence longer executions are encoded by using new variables for each time index.

Exercise 2 (Encode new plant dynamics). Consider the following plant dynamics:

$$x(t+1) = Ax(t) + Bu(t),$$

where

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

- Write the linear constraints encoding the above dynamics.